Assignment Day 2
C++ Programming

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1. Find roots of equations.

Program :

```
#include<iostream>
using namespace std;
// function to find roots
void rootofequation (int a, int b , int c){
   double discriminant, root1, root2, real, imaginary;

   discriminant = b*b - 4*a*c;
   // two different roots when discriminant > 0
   if (discriminant > 0 ){
       root1 = (-b + sqrt(discriminant)) / (2*a);
       root2 = (-b - sqrt(discriminant)) / (2*a);

       cout << "Roots are real and different" << endl;
       cout << "Root 1 = "<<root1 << endl;
       cout << "Root 2 = "<<root2 << endl;
   }
   // equal roots(Same)
   else if (discriminant == 0){
       root1 = -b / (2*a);
       cout << "Roots are real and equal " << endl;
       cout << "Root1 = "<<root1 << endl;
       cout << "Root2 = "<<root2 << endl;
   }
   // complex(imaginary roots)
   else{
       real = -b /(2*a);
```

```
        imaginary = sqrt(-discriminant) / (2*a) ;

        cout << "Roots are complex" << endl;
        cout << "Root1 = " << real << " + " << imaginary << "i"
<<endl;
        cout << "Root2 = " << real << " - " << imaginary << "i"
<<endl;
    }

}

int main (){

    int a,b,c;
    cout << "Enter coefficients a,b,c for equation (ax^2 + bx + c)
: " ;
    cin >> a >> b >> c;
    // function calling
    rootofequation( a, b, c);

    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./rootsofequation.o
Enter coefficients a,b,c for equation (ax^2 + bx + c) : 20
8
7
Roots are complex
Root1 = 0 + 0.556776i
Root2 = 0 - 0.556776i
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./rootsofequation.o
Enter coefficients a,b,c for equation (ax^2 + bx + c) : 1 5 6
Roots are real and different
Root 1 = -2
Root 2 = -3
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

2. Check given number is odd or even number.

Program :

```
#include <iostream>
using namespace std;

// function to check odd or even
void check (int num){
   if ( num % 2 == 0)
       cout << "Given number is EVEN";
   else
       cout << "Given number is ODD";
}

int main(){
   int num;
   cout << "Enter a number to check : " << endl;
   cin >> num;
   // function call
   check(num);
   return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./oddEven.o
Enter a number to check :
20
Given number is EVEN%
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./oddEven.o
Enter a number to check :
11
Given number is ODD%
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

3. Check if the entered char is small letter or capital letter.

Program :

```
#include<iostream>
using namespace std;

// function to check the character is small or capital
void check(char c){
```

```
    if (c >= 'A' && c <= 'Z')
        cout << "Entered charachter is CAPITAL LETTER " << endl;
    else if ( c >= 'a' && c <= 'z')
        cout << "Entered charachter is SMALL LETTER " << endl;
    else
        cout << "not an alphabet" ;
}

int main (){
    char c;
    cout << "Enter a charachter : ";
    cin >> c;

    // funtion call
    check(c);
    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./smallCapital.o
Enter a charachter : a
Entered charachter is SMALL LETTER
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./smallCapital.o
Enter a charachter : A
Entered charachter is CAPITAL LETTER
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

4. Write a recursive function to obtain the running sum of n numbers.

Program :

```
#include<iostream>
using namespace std;

// recursive function to add sum of n numbers
int sumofnumber(int n){
    // base case
    if (n == 0)
        return 0;
    // recursive call
    else
```

```
        return n + sumofnumber(n - 1);
}

int main(){
    int n , sum;
    cout << "Enter a number : ";
    cin >> n;
    sum = sumofnumber(n);
    cout << "Sum of " << n << "Number is = " << sum <<endl;
    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./sumofnumber.o
Enter a number : 8
Sum of 8Number is = 36
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

5. Write functions for calculating area of circle, rectangle, and square and call these functions from the main function.

Program :

```
#include<iostream>
using namespace std;

// Funtion to calcualte area of RECTANGLE
int areaRectangle (int len, int breadth){
    return len * breadth;
}
// Funtion to calcualte area of SQUARE
int areaSquare (int side){
    return side * side;
}
// Funtion to calcualte area of CIRCLE
double areaCircle (int radius){
    return 3.14 * radius * radius;
}

int main (){
```

```
    int len, breadth, side, radius;

    cout << "Enter length and breadth for RECTANGLE : " << endl;
    cin >> len >> breadth;

    cout << "Enter side of SQUARE : " << endl;
    cin >> side;

    cout << "Enter radius of CIRCLE : " << endl;
    cin >> radius;

    // Function call
    cout << "AREA OF RECTANGLE : " << areaRectangle(len, breadth)
<< endl << endl;
    cout << "AREA OF SQUARE : " << areaSquare(side) << endl <<
endl;
    cout << "AREA OF CIRCLE : " << areaCircle(radius) << endl <<
endl;

    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./area.o
Enter length and breadth for RECTANGLE :
10 20
Enter side of SQUARE :
5
Enter radius of CIRCLE :
7
AREA OF RECTANGLE : 200

AREA OF SQUARE : 25

AREA OF CIRCLE : 153.86

(base) abhinyaay@Abhinays-MacBook-Air day2 %
```

6. Write a function for factorial and use the function for printing factorial series like 1 2 6 24 120.....

Program :

```cpp
#include <iostream>
using namespace std;

// Function to calculate factorial

int factorial (int n){
   int fac = 1;
   for (int i = 1 ; i <= n ; i++){
       fac = fac * i ;
   }
   return fac;

}

int main (){
   int num;

   cout << "Enter number for series :  ";
   cin >> num;

   // series printing
   for (int i = 1; i <= num ; i++){
       // function call
       cout << factorial(i) << "  ";
   }

   return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air day2 % ./factorialSeries.o
Enter number for series :  8
1  2  6  24  120  720  5040  40320  %
(base) abhinyaay@Abhinays-MacBook-Air day2 %
```